



UKCP09: Spatially coherent projections

UKCP09 additional product



<http://ukclimateprojections.defra.gov.uk>

Spatially coherent projections: Executive summary

- The probabilistic climate projections available from UKCP09 website are not fully coherent across different locations. To further support those using UKCP09 who need to consider climate change at more than one location in a way that captures the relationship between the different locations, we have generated an 11-member ensemble of Spatially Coherent Projections (SCPs).
- The SCPs are not a replacement for UKCP09.
- The SCPs allow users requiring spatial coherence to explore the nature of projected impacts and vulnerabilities over 11 plausible snapshots of climate change for the UK during this century. Once identified, the full set of UKCP09 probabilistic projections can be used to estimate the likelihood of realising the associated impacts and vulnerabilities.
- The SCPs are based on the ensemble of regional climate models (RCMs) that were used in the construction of UKCP09. The SCPs are generated by applying scaling factors to the RCM data, so that the changes in the SCP ensemble that are linearly related to changes in global temperature better reflect the changes in global temperature consistent with the wider set of uncertainties explored by UKCP09, rather than those associated with the 11 GCMs used to drive the RCMs. Therefore, whilst the SCPs undersample the uncertainties in UKCP09, they explore a wider range of climate change than the RCM ensemble.

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1 Background

UKCP09 provides the first probabilistic assessment of climate change over the UK for the 21st century. It is radically different from its predecessor, UKCIP02, which consisted of a few 'plausible snapshots' of what UK climate might look like at various periods of the 21st century under four different emissions scenarios, and which contained very limited information on uncertainty. In contrast, UKCP09 includes much more comprehensive information on uncertainty and therefore can better support decision making. It does this by providing planners and climate impact scientists with 10,000 plausible realisations of how several climate variables might change over the coming century, *for a given location and emission scenario*. These realisations are based on plausible variants of the Hadley centre model, but also account for structural uncertainty inferred from models from other climate centres.

Often users must consider multiple variables (where multiple variables includes multiple climate variables, or multiple meaning periods, e.g. monthly or seasonal means, multiple time periods, multiple locations, or a combination of these). Changes in different variables are generally related to each other, and it is often crucial to account for the coherency between the multiple variables to make a meaningful decision.

In UKCP09, the issue of what variables can and cannot be treated as coherent is complex. On the one hand, the climate model data underpinning the probabilistic projections are fully coherent in time and space, as the processes in the climate system represented by the climate model impart a spatial and temporal coherence on the data. Despite this, the UKCP09 probabilistic climate projections are not fully spatially coherent. This is because the conversion of a limited ensemble of climate model simulations into probabilistic projections requires the use of statistical emulators, trained on the climate model results, to estimate the results of a much larger sample of plausible projections for multiple variables and locations. The emulated results are approximate, and therefore carry multivariate correlated errors which need to be accounted for if the results are to be fully coherent across different variables and locations. In principle, a full multivariate calculation is possible; however, limitations in the computing power and statistical techniques available at the time of production meant that climate variables in UKCP09 had to be processed in smaller subsets. The result was that full coherence could only be captured for climate variables within a given subset.

In UKCP09, we decided to address these limitations by splitting all the various combinations of climate variable, meaning period, time period and location. This was achieved in two ways. First, different spatial locations were processed

separately because many climate impacts models are configured to treat specific locations. For instance, the UKCP09 Weather Generator (Jones *et al.* 2009) is site specific, as are many crop models. Second, climate variables had to be split into two different sets, termed 'batch 1' and 'batch 2' in Annex 4 of Murphy *et al.* (2009). In each batch, full coherence was achieved between projected values for different climate variables, times of year, and future periods, *at any given specific location*. However, in general it was not possible to account for correlations between statistical errors at different locations, or between errors for climate variables in different batches.

In some circumstances, it is possible to bypass this lack of spatial coherence in UKCP09, by analysing these multiple variables one-by-one, effectively treating the variables as independent of each other, and then pooling the results. For instance, it is meaningful to use UKCP09 probabilistic projections point-by-point to produce a map of some relevant statistic or climate impact. The probability maps available from the UKCP09 website are an example of this and should be regarded as a graphical representation of a set of independent analyses. This is valid as long as the changes or impacts implied at different grid points in the probability maps are considered separately and are not treated coherently. This means that they should not be regarded as outcomes that would necessarily happen together in the observed future evolution of 21st century climate. Another example is the *wider range of uncertainty* in the Key findings (see Section 4.2.2 of Murphy *et al.* 2009) which are defined as the lowest and highest values in the 10 and 90% probability maps across all three emission scenarios.

Due to the lack of spatial coherence, it is not possible to use the UKCP09 probabilistic climate projections to inform decisions which need to consider multiple locations in a way where the locations cannot be treated separately. For instance, some national assessments require this. Another example is the assessment of implications of large scale water diversions (e.g. a pipeline). In contrast, UKCIP02 projections, which were derived from RCM runs that are spatially and temporally coherent for all variables, could be used to coherently assess climate change at multiple locations, albeit with no ability to explore potential decisions or policy options in the context of a range of projected climates.

In UKCP09, most of the uncertainty was determined from GCM runs done at about 300 km resolution. But like UKCIP02, information at the required fine resolution (50 km for UKCIP02, 25 km for UKCP09) was provided by running a small ensemble of regional climate model (RCM) runs. In UKCP09, this was 11 RCMs (see the UKCP09 guidance on 11-member RCMs) driven by 11 of the GCM runs that explored uncertainty in atmospheric and land processes. The main advantage of the RCMs, is that they provide spatially and temporally coherent simulations of future climate that include how the different variables relate to each other. The main disadvantages of the RCM simulations are that unlike the probabilistic projections, the RCM ensemble does not sample structural uncertainty in the atmospheric processes using alternative climate models, does not explore uncertainty arising from the carbon cycle, sulphur cycle, and ocean physics, and is only run for the A1B (medium) emissions scenario. Section 5.3 of Murphy *et al.* (2009) describes in more detail the pros and cons of the RCM ensemble compared to the UKCP09 probabilistic projections.

Here we aim to provide a set of Spatially Coherent Projections (SCPs) that satisfy several requirements:

- there is a set of SCPs for all three UKCP09 emission scenarios
- the ensemble size of the SCPs is 'small' to satisfy a user requirement that the SCPs are easy to use in decision making where many variables are being considered. We set the ensemble size of the SCPs to 11, the same size as the RCM ensemble.
- the SCPs are based on the spatially coherent RCM data from UKCP09, but are more consistent with the uncertainties represented in the UKCP09 probabilistic climate projections (here referred to as a more consistent spread).

The last requirement is done by applying scaling factors to members of the RCM ensemble, where the scaling factors incorporate sources of uncertainty accounted for in UKCP09 but not in the RCM ensemble. We only account for the spread in response associated with the enhanced spread of global temperature changes in UKCP09 relative to those changes in global temperature explored by the RCM ensemble; information from other climate models is only included through their impact on the uncertainty in changes in global temperature. To include all sources of uncertainty to provide a set of SCPs fully consistent with the probabilistic climate projections remains a long-term research goal that requires improved multivariate statistical techniques and greater computational power.

2 Method

Here, we propose a time scaling method (e.g. Mitchell, 2003; Harris *et al.* 2006; Murphy *et al.* 2009) to incorporate the uncertainty in global temperature from emission scenario, carbon cycle, sulphur cycle and ocean physics, into the RCM data, thereby making the spread of the scaled RCM data somewhat more representative of the overall spread in the probabilistic projections, whilst preserving spatial and temporal coherence.

For any climate variable and meaning period, e.g. monthly or seasonal mean, the anomaly relative to the 1961–1990 modelled baseline from the i^{th} RCM run, $R_i(\underline{x}, t)$, over points $\underline{x}$ (25 km grid boxes or administrative region) and at time t is decomposed into a component that is linearly related to global temperature change, $\beta_i(\underline{x})\Delta T_i(t)$, and a residual component, $\varepsilon_i(\underline{x}, t)$, such that

$$R_i(\underline{x}, t) = \beta_i(\underline{x})\Delta T_i(t) + \varepsilon_i(\underline{x}, t) \quad (1)$$

In UKCP09, realisations of time-dependent changes in global mean temperature were obtained by sampling globally averaged values of climate sensitivity, ocean heat uptake, sulphate aerosol forcing, and carbon cycle feedback, and then using a simple climate model (SCM) driven by time series of imposed changes in radiative forcing to convert these into plausible time series of values of $\Delta T(t)$ (see Section 3.2.4 of Murphy *et al.* 2009). In Equation 1, $\Delta T_i(t)$ is the time series of global temperature change from a configuration of the Simple Climate Model (SCM) (see Section 3.2.4 of Murphy *et al.* 2009) tuned to fit the model output from the GCM run used to drive the i^{th} RCM run. Then $\beta_i(\underline{x})$ is a spatial pattern of coefficients for the i^{th} RCM run used to scale the time series of global temperature change to provide the component that is linearly related to global temperature change. The spatial pattern of coefficients, $\beta_i(\underline{x})$ and the residuals, $\varepsilon_i(\underline{x}, t)$, are estimated using ordinary least squares regression for each climate variable, meaning period, and RCM run.

We can then replace the SCM that represents the GCM used to drive the i^{th} RCM with one that samples a different configuration of the carbon cycle, ocean diffusivity, and aerosol forcing. The SCM can even be forced by a set of emissions that are different to the ones used in the GCM run, thus allowing SCPs to be extended to other emission scenarios. The time series of global temperature from this alternative SCM, $\Delta T_i^{\text{new}}(t)$, is then used to produce a spatially coherent projection (SCP) for which the anomalies relative to 1961–1990 baseline are

$$R_i^{\text{new}}(\underline{x}, t) = \beta_i(\underline{x})\Delta T_i^{\text{new}}(t) + \varepsilon_i(\underline{x}, t) \quad (2)$$

By maintaining the same pattern of scaling coefficients, $\beta_i(\underline{x})$, and preserving the residual component and all correlations that they capture, we retain temporal and spatial coherence. To demonstrate this more clearly it is simple to show that the difference between the SCP and original RCM runs is just the scaled difference of the two time series of global mean temperature:

$$R_i^{\text{new}}(\underline{x}, t) - R_i(\underline{x}, t) = \beta_i(\underline{x})[\Delta T_i^{\text{new}}(t) - \Delta T_i(t)] \quad (3)$$

As the ensemble mean of the N_{rcm} SCPs is:

$$\frac{1}{N_{\text{rcm}}} \sum_{i=1}^{N_{\text{rcm}}} [\beta_i(\underline{x}) \Delta T_i^{\text{new}}(t) + \frac{1}{N_{\text{rcm}}} \sum_{i=1}^{N_{\text{rcm}}} \varepsilon_i(\underline{x}, t)], \quad \text{Equation 4}$$

the second term on the right hand side of Equation 4 contains the ensemble mean of the residual terms. This captures the mean regional climate change signal not explained by global mean temperature change, which consists of a mixture of regional effects in the global climate model simulation used to drive the RCMs, plus the mean impact of local effects captured by the RCMs themselves.

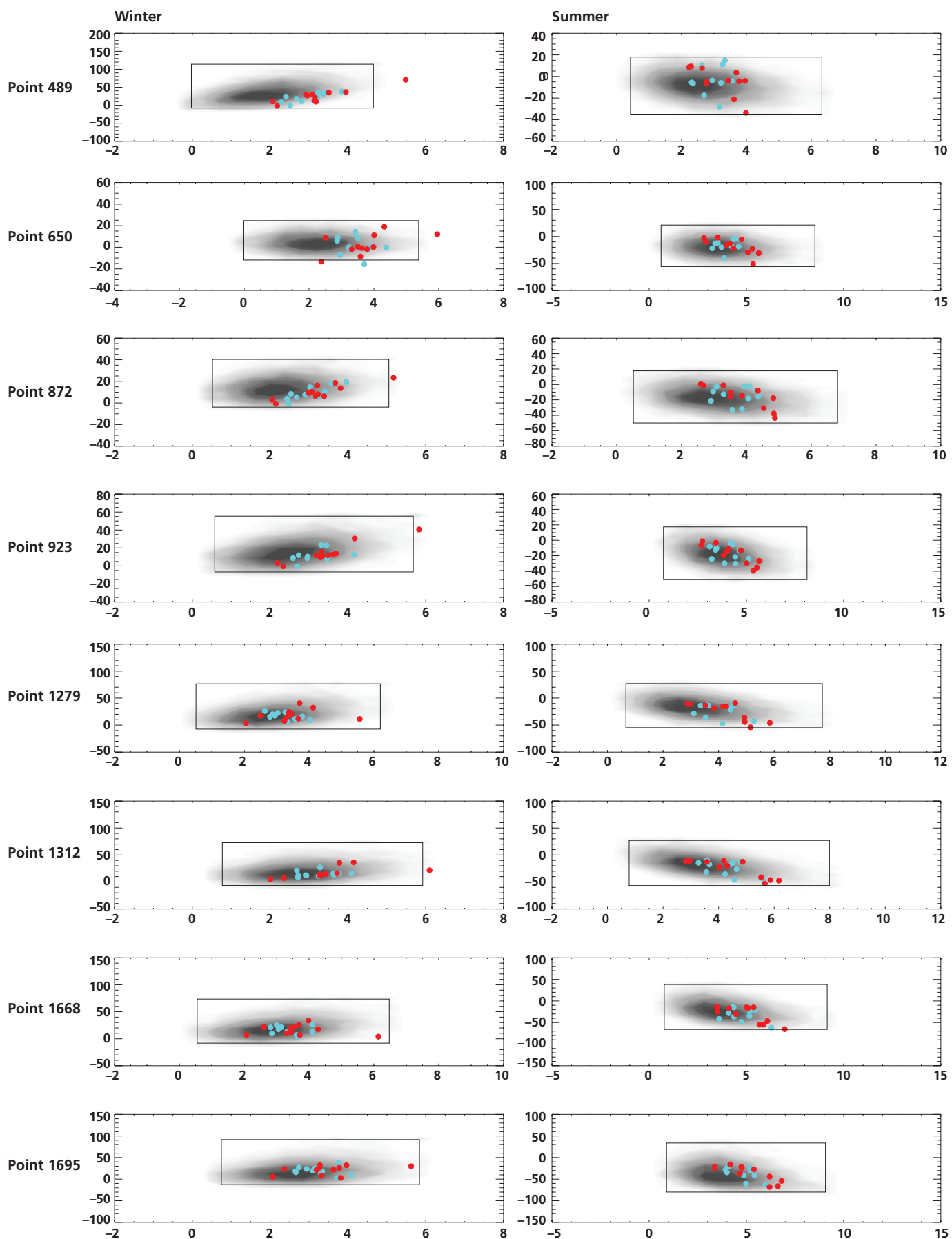
From UKCP09, for each of the 10,000 model variants used to make the sampled data, and for each emission scenario, we have an SCM. We use Equation 2 to combine each RCM with one configuration of the SCM sampled from the set of 10,000 SCMs for that emission scenario. This process generates an ensemble of 11 SCPs that better reflect the uncertainty in the probabilistic climate projections related to global temperature. However, some choices of 11 SCMs are going to be more representative of the probabilistic climate projections than others, and will allow users to explore a wider range of plausible future climates that are more consistent with the UKCP09 probabilistic projections. To find a good choice of 11 SCMs, we try hundreds of randomly selected sets of 11 SCMs and choose the one that maximises a cost function, which is described in the next Section.

3 Results

The previous Section described how we make an 11-member ensemble of spatially coherent projections (SCPs). Here we describe the process of building a large sample of candidate SCP ensembles and picking the one that maximises a cost function which measures some property of each candidate SCP ensemble. The main thing we want to achieve is a set of SCPs that more effectively explore the uncertainties in the full set of UKCP09 probabilistic projections than the original RCM ensemble members for all the different combinations of climate variables, locations, time periods, and meaning periods. In practice, we focus on 4 variables: winter and summer changes in temperature and precipitation for all 25 km grid boxes over the UK for the period 2070–2099. For each candidate SCP ensemble, we measure total spread across all locations for these four variables using the sum of the Mahalanobis distance (a multivariate measure of distance) (Mardia *et al.* 1997), over all pairs of the 11 members of the SCP ensemble.

To illustrate some of the issues with generation of an SCP ensemble, Figure 1 shows a candidate SCP (red dots) and compares it with raw data from the 11-member RCM ensemble (light blue dots) and the sampled data from UKCP09 for these 4 variables at eight locations in the UK (see Figure 2). In this particular sample, several data points in this possible SCP ensemble have gone outside the 1st–99th credible range* (the bounding box in Figure 1) for some variables. That some data points go outside the 1st–99th credible range is statistically inevitable; on the other hand, use of values outside the 1st–99th credible range is not recommended, according to the UKCP09 guidance*, but providing such values that go outside the 1st–99th credible range goes against the guidance in UKCP09. So really we want to achieve a balance between an enhanced spread in the SCP ensemble relative to the RCM ensemble whilst minimising the number of times that SCP data go outside the 1st–99th credible range. Therefore, we define an empirical cost function, $COST = SPREAD - PENALTY$ that reflects this balance, where *SPREAD* is the measure of total spread discussed above and *PENALTY* is a factor that ranges steadily from 1.0 (if there are no SCP data outside the 1st–99th credible interval for all four variables and locations) to 0.0 if there are on average 1 or more of the 11 SCP points outside the 1st–99th credible range across the four variables and all

* In UKCP09, guidance is provided on the robustness of the different probability levels. The 10–90% credible range, which is the values between the 10 and 90% probability level, is considered to be robust, the 1–9% and 91–99% credible ranges are considered to be useable but with caution, and values below the 1% probability level or above the 99% probability level are not recommended for use. For the sampled data, values below the 1% probability level were set to the value of the 1% probability level, and values above the 99% probability level were set to the value at the 99% probability level. Here we use the credible range consistent with the treatment of the sampled data.



locations. Then we repeatedly produce candidate 11-member SCPs 50,000 times, and pick the 11-member SCP that maximises the cost function. Figure 3 shows for these example locations that iteration 571 happened to maximise the cost function. Even with this cost function, it is not guaranteed to have all SCP data lie within the UKCP09 credible ranges, as the cost function is designed to find a SCP ensemble with the best balance between maximising the spread and minimising the number of SCP data outside the credible range.

It is clear from Figure 3 that the SCPs enhance the spread of sampled changes compared with the original RCM projections, for all four variables shown. That the SCP ensemble is more representative of the UKCP09 probability distributions is also clearly seen in Figure 4 which compares the SCP ensemble, RCM ensemble, and full UKCP09 probability distribution for changes in summer temperature for 2070–2099 relative to 1961–1990 at a handful of grid points. Furthermore, Figure 5 shows that the spread of the SCP is generally greater than that of the RCM ensemble for the four variables at all grid points around the UK by a factor of 1–2.

For some variables like summer temperature change, which is strongly related to changes in global temperature, our method has worked well at enhancing the spread of the RCMs. For variables like summer precipitation there is not much change in spread relative to the RCMs. It is even possible by chance to reduce the spread, e.g. for changes in winter precipitation over parts of Northern Ireland though this does not occur often. This reduction occurs because the change in spread for a given variable at a given location is related to the correlation across the ensemble of a) the scaling coefficients and b) the changes in global temperature. To maximise spread it requires the low scaling coefficients to be multiplied by the lowest global temperatures and the highest scaling coefficients to be multiplied by the highest global temperatures. So if the correlation of the scaling coefficients with the 11 original RCM global temperatures is bigger than the correlation with the new set of 11 global temperatures then the spread goes down. The cost function is designed to ensure this does not happen often, and Figure 5 is confirmation that the algorithm is largely successful.

However, it is also clear that the SCP data bears a closer resemblance to the RCM ensemble than the underlying UKCP09 sampled data, although they do not underestimate the overall uncertainty present in the full probabilistic climate projections as much as the RCM ensemble. This is because UKCP09 probabilistic climate projections accounted for uncertainty from other sources of information, mainly from the set of other climate models included in them. For instance, the SCP members fail to capture temperature changes less than 2°C in winter, and also fail to sample the possibility of a modest increase in summer rainfall over Southern England, in contrast to the UKCP09 sampled data. This is because the SCPs share the tendency of our perturbed physics ensemble to simulate changes in summer precipitation that are more negative than the other international climate models. Whereas UKCP09 was designed to capture information on regional scale from other models, the SCP methodology is not designed to capture this information because it cannot be done in a way that retains spatial coherence; as out-lined above, the only information from other models that is accounted for in the SCP ensemble is their affect on the changes in global temperature found in UKCP09. To make an SCP ensemble that is more representative of the probabilistic climate projections, requires a more sophisticated, powerful multivariate framework which is a long-term research goal. Therefore, the SCPs should be regarded as an interim product.

Figure 1 (see page 10): Comparison of SCP data (red) with RCM data (light blue) and UKCP09 sampled data (grey shading) for an example of a candidate 11-member SCP ensemble where several data lie outside the bounding boxes, which are generated from the 1 and 99% percentiles of changes in temperature and precipitation from UKCP09. Comparisons are made for eight locations (see Figure 2) and for changes in 2070–2099 relative to 1961–1990 in winter temperature (K) and precipitation (in %) (left hand side) and the corresponding changes in summer (right hand side). The grey shading is darkest where there is the greatest concentration of sampled data from UKCP09. The outer contour of the grey shading is the smallest area that contains 99% of the sampled data. Points 489–1695 are shown in Figure 2 with lower numbers indicating more northerly locations.

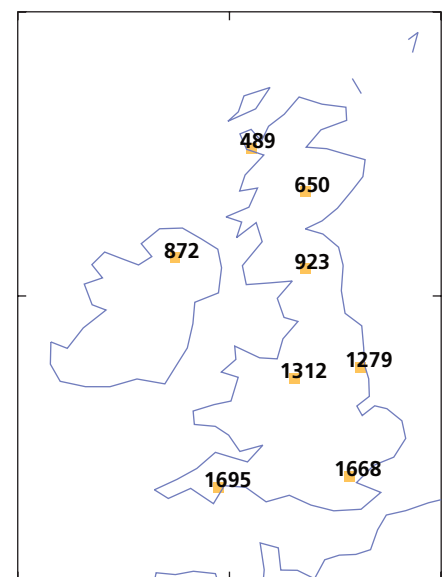


Figure 2: Location of the eight 25 km grid boxes used in Figures 1 and 3.

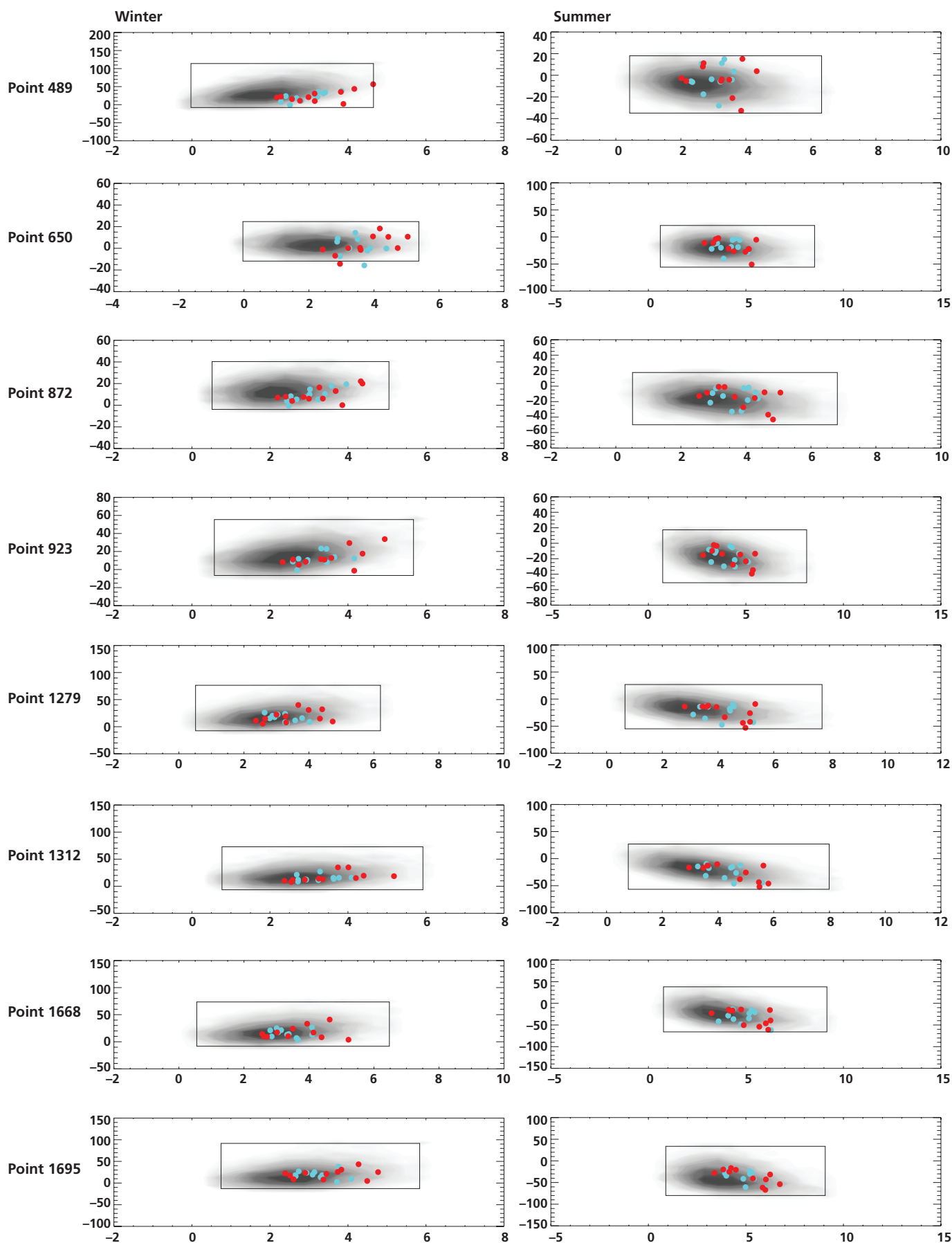


Figure 3: As Figure 1 but for the 11-member SCP that maximises the cost function and is used to make the SCPs for other variables.

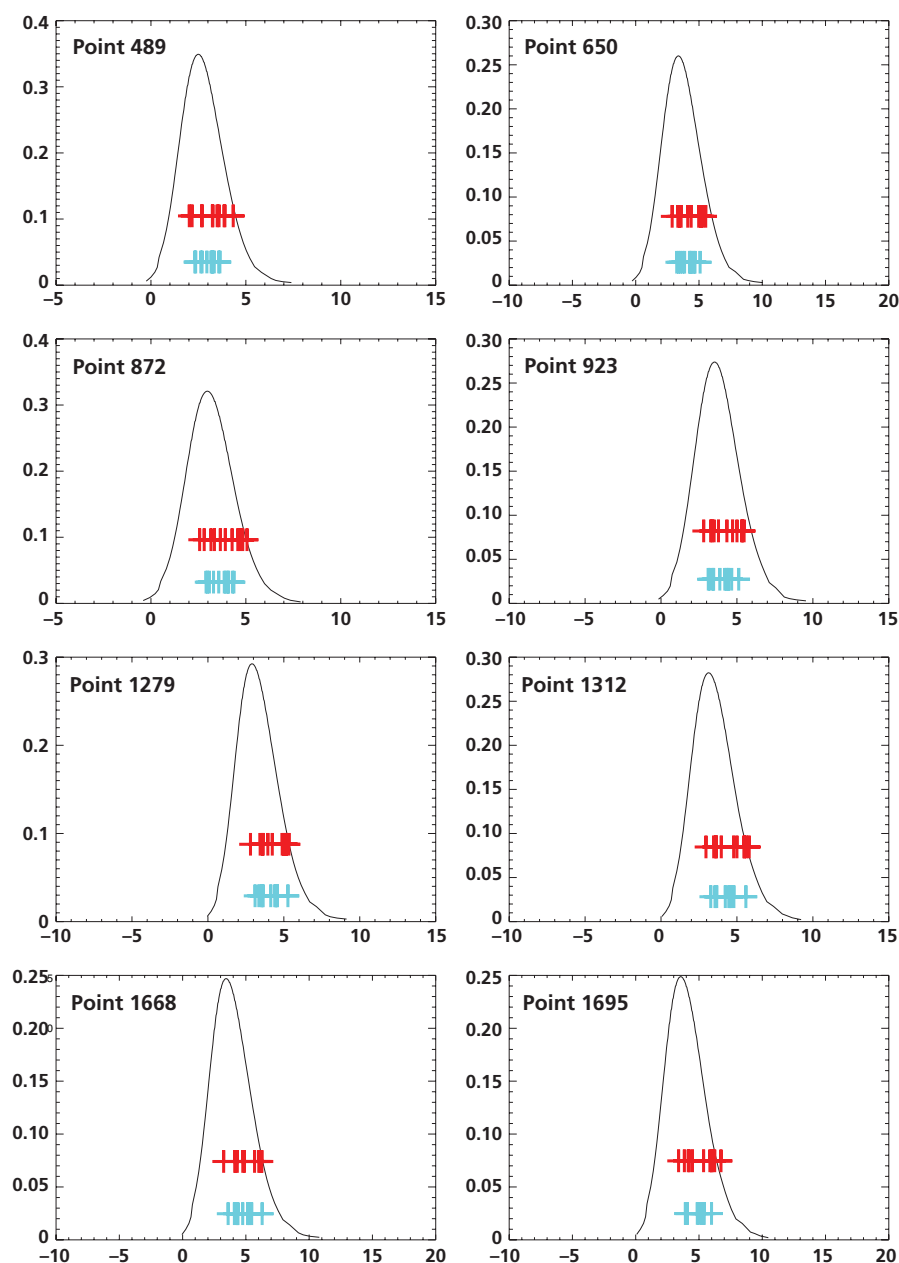


Figure 4: Comparison of SCP data (red crosses) with RCM data (light blue crosses), offset in along the y-axis for clarity, and the probability distribution from UKCP09 for the 11-member SCP ensemble comparisons are made for eight locations (see Figure 2) and for changes in 2070–2099 relative to 1961–1990 in summer temperature (K). Points 489–1695 are shown in Figure 2 with lower numbers indicating more northerly locations.

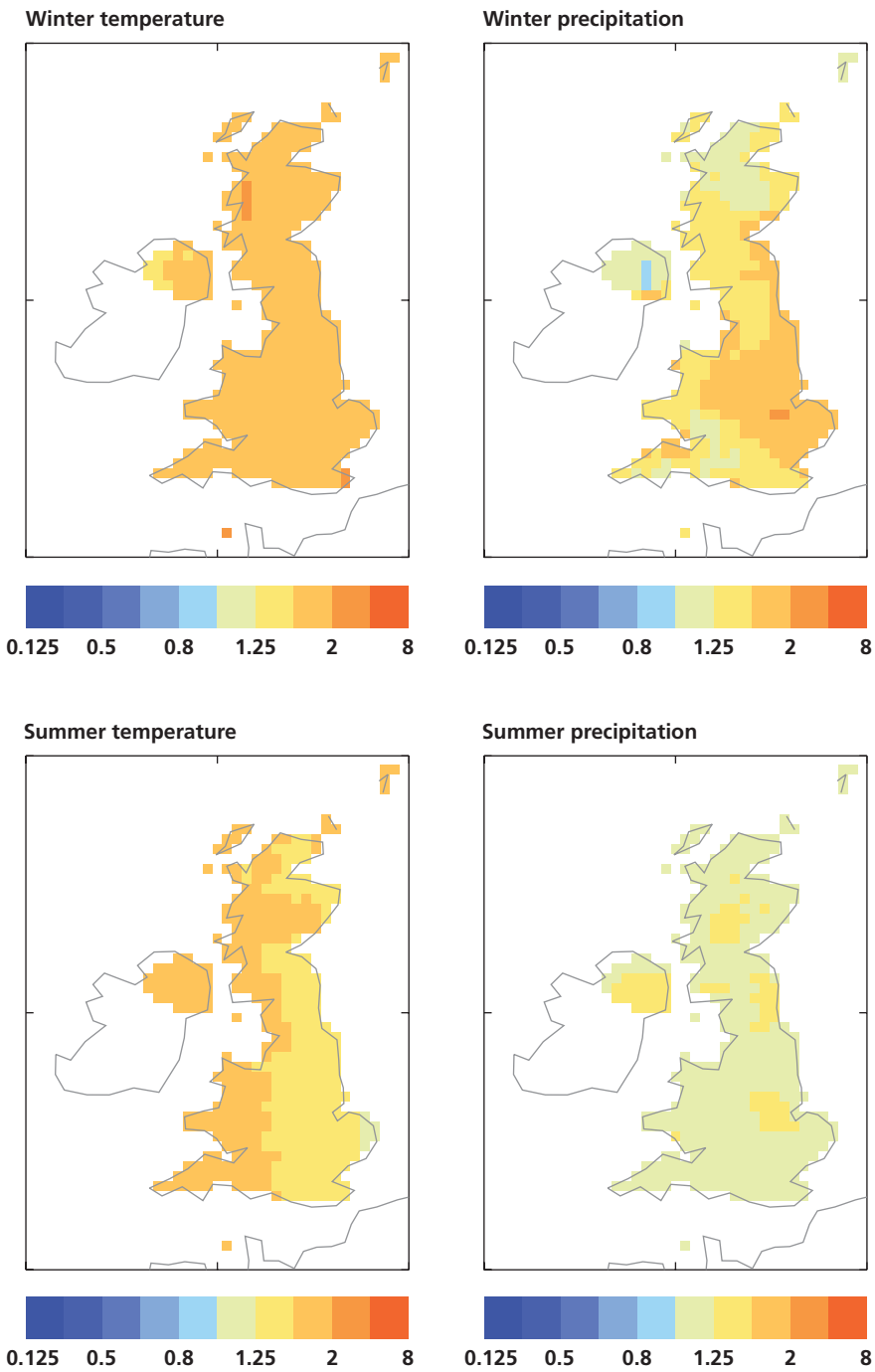


Figure 5: Ratio of standard deviation of SCP ensemble relative to RCM ensemble for summer and winter changes in temperature and precipitation.

4 Outputs

As an example of the actual output, Figure 6 shows an example of changes in winter and summer temperature and precipitation from the first member of the 11-member SCP ensemble. The patterns could have plausibly come from an RCM run.

The full SCP product will be delivered as a set of CSV-formatted text files within the UKCP09 CSV archive accessible from the User Interface. For each of the 11 members of the SCP ensemble and each variable, there is a file storing values in a two-dimensional grid; this arrangement is identical to that of the files obtained by users who download the probability maps in CSV format. SCPs are on the same 25 km grid as used by the regional climate models and UKCP09 probabilistic projections, and will be available for all climate variables (including windspeed), meaning periods, time periods, and emission scenarios for which there is already probabilistic data.*

* SCPs are based on raw RCM data which are fully coherent, so there is no need to split the data into batches as is done with the probabilistic information in UKCP09.

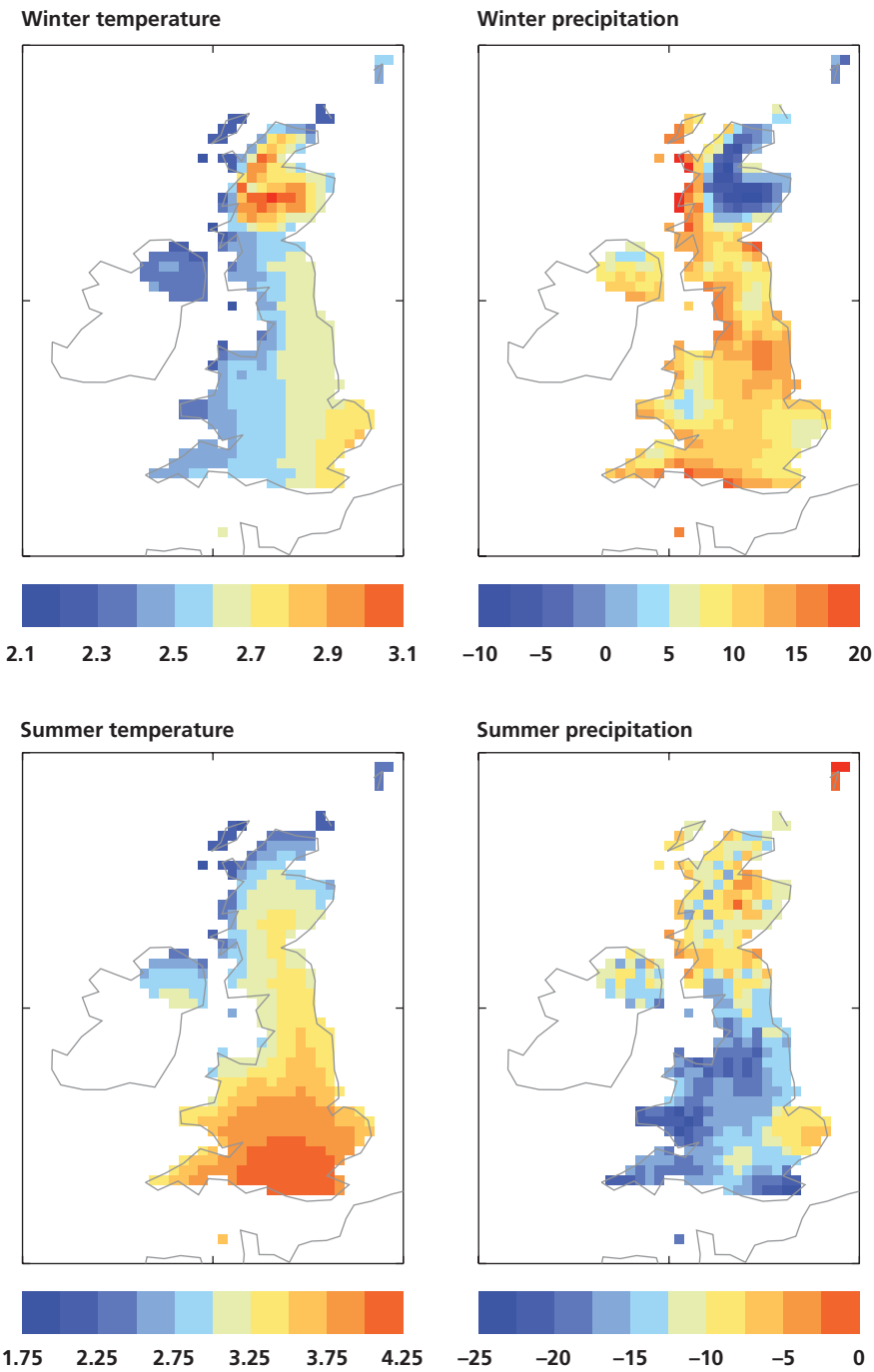


Figure 6: Example from the 1st member of the SCP ensemble of changes in winter and summer temperature (K) and precipitation (%) for 2070–2099 relative to 1961–1990.

5 Considerations in using SCPs

For a user of UKCP09, the first consideration is whether UKCP09 probabilistic projections can be used for the climate impact of interest. This climate impact has to be some function of a set of input variables where each input variable is any combination of climate variable, meaning period, e.g. seasonal or monthly mean, future time period, and location available in UKCP09. Coherence is required across this set of input variables that are used in a single evaluation of the climate impact. With the production of the SCPs, it means that users can now deal with problems where the UKCP09 probabilistic projections are not coherent. As coherence is required across spatial locations or different 'batches' (see Chapter 1), there are three clear examples of when the SCP ensemble should be used:

- a user needs to consider how two or more grid points vary together across different realisations, e.g. when the climate impact is calculated from the difference between North and SE England summer rainfall.
- a user is making a nationwide assessment where locations cannot be treated separately.
- a user needs to use variables that are in different UKCP09 batches as inputs to an impacts model

Two examples of where users might think they require spatial coherence but in fact do not are:

- producing maps at a specified probability level from sampled data. Users would loop through each grid point, take the 10,000 sampled points, order them and take the value that corresponds to the specified probability level, e.g. for 90% probability level the 9000th value would be used. Each grid point is treated separately but results are plotted together to make the map.
- running UKCP09 Weather Generator at every grid point and plotting a map of heaviest rainfall (although this is not very practical). Each call of the Weather Generator just needs to know the values of the driving variables at each grid point, and it does not matter what goes on at the other grid points. However, the driving variables at each grid point need to be coherent and in UKCP09 they are.

There will be situations where both the UKCP09 probabilistic projections and the SCPs could be used. As there are 10,000 plausible realisations to consider in the probabilistic projections, the small set of SCPs appears to be a more practical option. However, this means that the SCPs will under-sample the full set of outcomes implied by the UKCP09 probabilistic projections. Another limitation of the SCPs is that there is no information on the likelihood of that level of climate change being realised. Therefore in these situations where both the UKCP09 probabilistic projections and SCPs could be used, the UKCP09 probabilistic projections should be used.

Consequently, the SCPs are not a replacement for UKCP09 but are designed to complement the full probabilistic projections. As such, they should be treated with the same caveats as the raw RCM data (see UKCP09 online Guidance). For instance, the SCPs could be used for an initial assessment of vulnerability of a particular system/sector to climate change. Then it would be important to assess any identified vulnerabilities in the context of the probabilistic projections to determine their relative likelihood, to determine the time frame at which climate change becomes an issue that requires adaptation, to better understand the wider range of uncertainty, and thereby further inform the assessment of the nature and scope of the possible impacts and associated vulnerabilities and risks. It is worth noting that to these ends, the task has been made much easier through the introduction of maps of probability of a climate change variable being above or below a user-defined threshold.

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